Annex I. Public and External Sector Debt Sustainability Analysis

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Ghana Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgment	No

External and overall debt are at high risk of debt distress but remain sustainable. The shocks from COVID-19 epidemic (collapse of oil prices, decline in trade, and lower non-commodity growth) are expected to deepen current account and fiscal deficits over the medium term resulting in a higher debt path compared to the November 2019 DSA.

1. The end of exceptional spending items (COVID-19-related spending, financial sector restructuring) and the government's commitment to the fiscal rule underpin the downward path of debt from 2022. Ghana enters the crisis relatively well prepared. External buffers have been strengthened with better-than-expected outturn in 2019 due to higher gold export receipts and remittances, and a $3bn Eurobond issue in early 2020. Contingent risks have been alleviated thanks to progress on the financial sector clean-up and the launch of the energy sector reform program. Gold prices (Ghana's main export) boosted by the global flight to safety and the pre-sale of most of 2020 cocoa harvest mitigate the impact of the shock this year.

2. The standard stress tests have been augmented to reflect a possible scenario with a stronger outbreak and protracted national lock-down. The growth shock has been increased to two standard deviations and exchange rate depreciation has been increased to 40 percent. Under these extreme shocks, debt still remains sustainable. Debt-service indicators are not on an explosive path. Furthermore, the DSA shocks likely exaggerate the impact on these indicators over the medium-long run given that, once the COVID emergency is solved and the elections are over, stressors such as risk premia, low commodity prices, and weak domestic revenues would improve significantly. The inclusion of the contingent liability stress test at 5 percent of GDP should also be adequate to cover additional financial sector costs from the impact of COVID-19.

3. Nevertheless, even under the baseline, risks to the outlook remain important. Risks depend primarily on the depth and duration of the COVID-19 and oil price shock. Fiscal costs including pandemic response and measures to support economic activity could exceed those envisioned in the baseline and extend into 2021. A deeper global slowdown could have a greater impact on oil

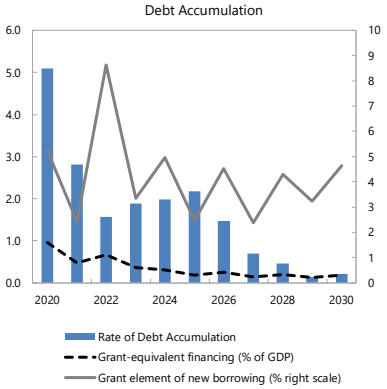
prices, private transfers and investment and further weaken the exchange rate. A more prolonged crisis could create additional liquidity risks into 2021. Stress tests show that, of these risks, exchange rate depreciation, export and commodity prices shocks are associated might have the greatest impact on debt sustainability.

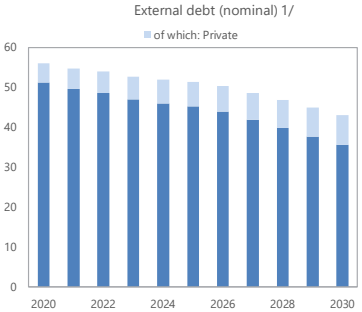
Table 1. Ghana: External Debt Sustainability Framework, Baseline Scenario, 2017-2040

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**43.8**	**45.4**	**47.0**	**56.0**	**54.7**	**53.9**	**52.6**	**51.9**	**51.4**	**43.1**	**35.4**	**35.3**	**50.4**
of which: public and publicly guaranteed (PPG)	**39.8**	**41.1**	**42.4**	**51.2**	**49.6**	**48.6**	**47.0**	**46.0**	**45.2**	**35.5**	**25.4**	**31.4**	**44.2**
Change in external debt	1.9	1.6	1.6	9.0	-1.3	-0.8	-1.4	-0.7	-0.5	-1.8	1.4		
Identified net debt-creating flows	**-5.5**	**-6.2**	**-4.2**	**-0.2**	**-5.1**	**-3.5**	**-5.4**	**-4.4**	**-4.2**	**-4.3**	**-2.9**	**-2.3**	**-4.2**
Non-interest current account deficit	**1.3**	**0.8**	**0.3**	**1.7**	**-0.3**	**-0.2**	**-0.1**	**-0.6**	**-0.3**	**-1.1**	**-0.5**	**4.3**	**-0.4**
Deficit in balance of goods and services	3.1	1.1	1.9	2.3	0.1	0.4	-1.2	-1.7	-1.5	-0.4	1.3	**6.7**	**-0.6**
Exports	34.6	34.4	34.5	34.1	35.2	34.3	34.4	33.5	32.3	27.0	18.4		
Imports	37.7	35.4	36.4	36.4	35.3	34.7	33.3	31.8	30.8	26.6	19.8		
Net current transfers (negative = inflow)	-4.2	-3.9	-5.1	-3.3	-3.2	-3.1	-2.9	-2.8	-2.7	-2.2	-1.4	**-4.2**	**-2.7**
of which: official	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	2.4	3.7	3.5	2.7	2.9	2.5	4.0	4.0	3.9	1.5	-0.5	**1.9**	**2.9**
Net FDI (negative = inflow)	**-5.5**	**-4.4**	**-4.9**	**-3.9**	**-5.0**	**-5.0**	**-5.0**	**-5.0**	**-5.0**	**-4.0**	**-3.0**	**-5.6**	**-4.8**
Endogenous debt dynamics 2/	**-1.3**	**-2.5**	**0.3**	**2.0**	**0.1**	**1.8**	**-0.2**	**1.1**	**1.1**	**0.8**	**0.6**		
Contribution from nominal interest rate	2.1	2.3	2.4	2.8	3.3	3.5	3.3	3.3	3.2	2.6	2.1		
Contribution from real GDP growth	-3.2	-2.5	-2.7	-0.7	-3.2	-1.7	-3.6	-2.2	-2.1	-1.8	-1.5		
Contribution from price and exchange rate changes	-0.2	-2.4	0.6	...	...	...	...	...	...	...	...		
Residual 3/	**7.4**	**7.8**	**5.8**	**9.2**	**3.9**	**2.7**	**4.0**	**3.7**	**3.7**	**2.5**	**4.3**	**4.4**	**3.8**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**35.7**	**42.7**	**44.0**	**43.0**	**41.4**	**40.9**	**40.4**	**31.6**	**23.0**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**103.5**	**125.4**	**125.1**	**125.4**	**120.2**	**121.9**	**124.9**	**117.2**	**124.7**		
PPG debt service-to-exports ratio	**4.8**	**5.0**	**14.3**	**19.4**	**15.3**	**15.9**	**14.7**	**14.5**	**15.8**	**18.8**	**19.1**		
PPG debt service-to-revenue ratio	**12.4**	**12.0**	**35.0**	**50.2**	**36.5**	**36.9**	**34.0**	**32.3**	**33.5**	**30.9**	**20.7**		
Gross external financing need (Million of U.S. dollars)	-965.0	-590.3	977.2	3602.4	893.2	1102.1	976.7	610.1	1122.1	2177.6	5734.9		
Key macroeconomic assumptions													
Real GDP growth (in percent)	8.1	6.3	6.1	1.5	5.9	3.3	7.1	4.4	4.3	4.3	4.7	**6.7**	**4.2**
GDP deflator in US dollar terms (change in percent)	-0.8	4.5	-3.3	-5.9	-2.3	2.5	1.3	1.6	2.3	2.3	2.1	**0.8**	**1.1**
Effective interest rate (percent) 4/	5.4	5.9	5.5	5.6	6.1	6.7	6.7	6.6	6.6	6.2	6.5	**4.3**	**6.4**
Growth of exports of G&S (US dollar terms, in percent)	17.0	10.2	10.2	-5.6	6.8	3.3	8.9	3.3	2.8	2.7	4.2	**13.6**	**3.1**
Growth of imports of G&S (US dollar terms, in percent)	8.4	4.3	5.3	-4.5	0.3	4.3	3.9	1.4	3.5	3.6	4.9	**9.2**	**2.4**
Grant element of new public sector borrowing (in percent)	...	...	...	5.2	2.4	8.6	3.3	5.0	2.5	4.6	0.0	**...**	**4.2**
Government revenues (excluding grants, in percent of GDP)	13.3	14.2	14.1	13.2	14.8	14.8	14.9	15.1	15.2	16.4	17.0	**12.9**	**15.2**
Aid flows (in Million of US dollars) 5/	352.7	180.3	191.0	394.9	446.3	930.5	462.4	535.7	341.3	567.2	2.1		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	1.0	0.5	0.7	0.4	0.3	0.2	0.2	0.0	**...**	**0.4**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	8.3	8.0	13.7	7.7	7.0	3.9	5.1	0.0	**...**	**6.5**
Nominal GDP (Million of US dollars)	58,978	65,518	67,240	64,183	66,361	70,316	76,300	80,900	86,320	119,006	230,456		
Nominal dollar GDP growth	7.3	11.1	2.6	-4.5	3.4	6.0	8.5	6.0	6.7	6.7	6.9	**7.7**	**5.4**
Memorandum items:													
PV of external debt 7/	...	...	40.2	47.5	49.1	48.4	47.0	46.8	46.5	39.1	33.0		
In percent of exports	...	...	116.7	139.5	139.6	141.0	136.5	139.5	144.0	145.2	178.9		
Total external debt service-to-exports ratio	7.5	8.0	17.5	22.9	18.9	19.7	18.7	18.8	20.4	25.6	32.4		
PV of PPG external debt (in Million of US dollars)			23982.4	27,403.65	29204.7	30246.6	31571.0	33078.4	34841.3	37588.4	53007.1		
(PVt-PVt-1)/GDPt-1 (in percent)				5.1	2.8	1.6	1.9	2.0	2.2	0.2	3.1		
Non-interest current account deficit that stabilizes debt ratio	-0.6	-0.8	-1.3	-7.3	1.0	0.6	1.2	0.1	0.2	0.7	-1.9		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

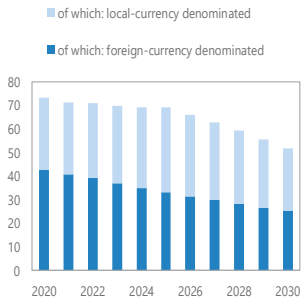
Table 2. Ghana: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017-2040

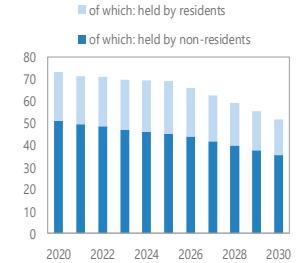
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**57.6**	**62.5**	**63.5**	**73.3**	**71.2**	**71.0**	**69.6**	**69.3**	**69.0**	**51.7**	**30.7**	**48.5**	**65.3**
of which: external debt	**39.8**	**41.1**	**42.4**	**51.2**	**49.6**	**48.6**	**47.0**	**46.0**	**45.2**	**35.5**	**25.4**	**31.4**	**44.2**
Change in public sector debt	0.0	4.9	1.0	9.8	-2.1	-0.2	-1.4	-0.3	-0.2	-3.7	1.9		
Identified debt-creating flows	**-2.6**	**0.7**	**2.9**	**8.5**	**-1.4**	**0.7**	**-0.5**	**0.4**	**0.5**	**-3.1**	**2.0**	**-0.4**	**-0.5**
Primary deficit	**-0.5**	**1.4**	**1.8**	**4.1**	**0.0**	**-0.5**	**-0.1**	**0.1**	**0.4**	**-2.8**	**2.2**	**2.1**	**-0.8**
Revenue and grants	13.9	14.5	14.3	13.5	15.1	15.1	15.1	15.2	15.2	16.4	17.0	**13.7**	**15.4**
of which: grants	0.6	0.3	0.3	0.4	0.3	0.3	0.2	0.1	0.1	0.0	0.0		
Primary (noninterest) expenditure	13.4	15.9	16.1	17.7	15.1	14.6	15.0	15.3	15.6	13.6	19.2	**15.8**	**14.6**
Automatic debt dynamics	**-2.0**	**-0.1**	**1.2**	**4.3**	**-1.4**	**1.2**	**-0.4**	**0.3**	**0.1**	**-0.3**	**-0.2**		
Contribution from interest rate/growth differential	-1.1	-0.5	-1.2	1.3	-0.8	1.5	-1.0	0.5	0.2	-0.2	-0.2		
of which: contribution from average real interest rate	3.2	2.8	2.4	2.2	3.3	3.8	3.8	3.4	3.1	2.1	1.1		
of which: contribution from real GDP growth	-4.3	-3.4	-3.6	-0.9	-4.1	-2.3	-4.7	-2.9	-2.8	-2.3	-1.3		
Contribution from real exchange rate depreciation	-0.9	0.4	2.4	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**-0.5**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.1**	**0.0**
Privatization receipts (negative)	0.0	-0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**2.5**	**4.1**	**-1.9**	**4.3**	**-1.3**	**-1.2**	**-0.4**	**-0.9**	**-0.8**	**-0.6**	**-0.1**	**3.5**	**-0.4**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**59.3**	**68.3**	**66.6**	**66.4**	**65.2**	**65.0**	**65.0**	**48.4**	**28.8**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**413.6**	**504.3**	**440.8**	**440.9**	**432.1**	**428.3**	**426.1**	**295.3**	**169.4**		
Debt service-to-revenue and grants ratio 3/	**100.8**	**68.4**	**91.7**	**106.8**	**79.4**	**72.7**	**79.8**	**81.2**	**76.1**	**61.2**	**33.0**		
Gross financing need 4/	13.5	10.8	14.9	18.6	12.0	10.5	12.0	12.4	12.0	7.2	7.8		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	8.1	6.3	6.1	1.5	5.9	3.3	7.1	4.4	4.3	4.3	4.7	**6.7**	**4.2**
Average nominal interest rate on external debt (in percent)	5.8	6.1	5.8	6.1	6.8	7.1	7.0	7.0	6.9	6.6	7.2	**4.4**	**6.7**
Average real interest rate on domestic debt (in percent)	8.8	7.1	4.1	1.3	4.5	6.3	6.6	5.3	4.3	3.3	2.3	**7.6**	**4.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.9	1.5	7.3	...	...	...	...	...	...	...	...	**2.5**	**...**
Inflation rate (GDP deflator, in percent)	10.4	10.2	8.8	9.3	8.3	7.3	6.4	6.5	6.4	6.3	6.1	**14.1**	**7.0**
Growth of real primary spending (deflated by GDP deflator, in percent)	-5.4	25.8	7.8	11.4	-9.4	-0.3	10.2	6.1	6.8	5.6	6.5	**8.7**	**2.9**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-0.5	-3.5	0.8	-5.6	2.1	-0.3	1.3	0.4	0.6	0.9	0.3	**-1.1**	**0.3**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

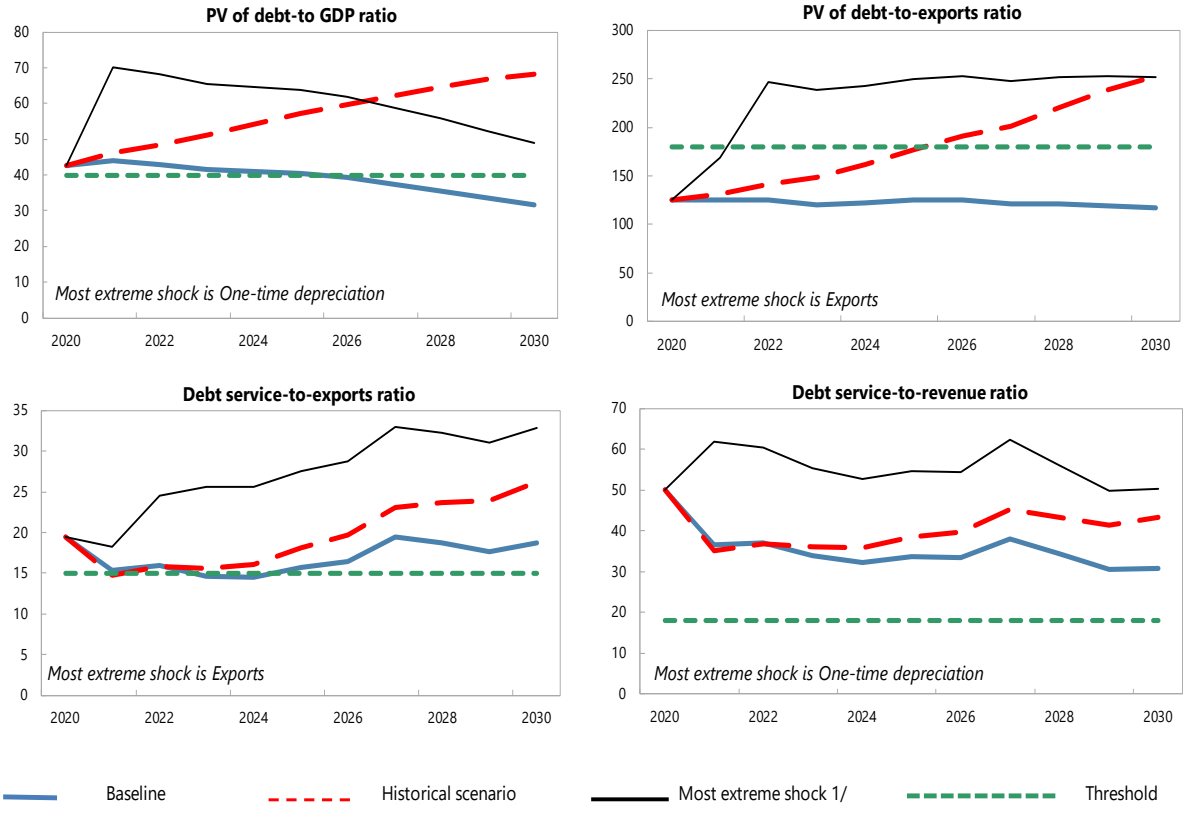
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Ghana: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2020-2030



Customization of Default Settings	Size	Interactions
Standardized Tests	Yes	Yes
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	8.5%	7.5%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	12	20
Avg. grace period	6	17

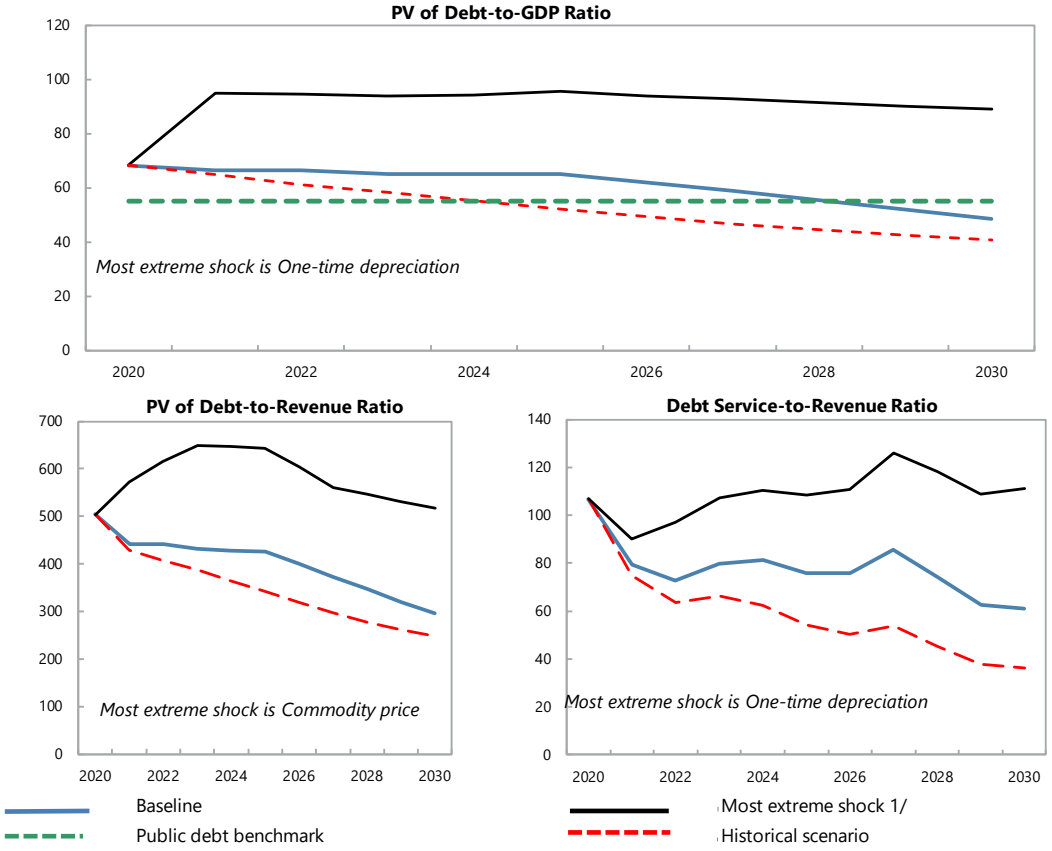
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Ghana: Indicators of Public Debt Under Alternative Scenarios, 2020-2030



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	44%	44%
Domestic medium and long-term	36%	36%
Domestic short-term	20%	20%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	8.5%	8.0%
Avg. maturity (incl. grace period)	12	20
Avg. grace period	6	17
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.1%	4.0%
Avg. maturity (incl. grace period)	5	7
Avg. grace period	4	6
Domestic short-term debt		
Avg. real interest rate	1.4%	2.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Ghana: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020-2030

(In percent)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	**43**	**44**	**43**	**41**	**41**	**40**	39	37	36	33	32
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	**43**	**46**	**48**	**51**	**54**	**57**	**60**	**62**	**64**	**67**	**68**
B. Bound Tests											
B1. Real GDP growth	**43**	**47**	**49**	**47**	**47**	**46**	**45**	**43**	**41**	38	36
B2. Primary balance	**43**	**46**	**49**	**48**	**47**	**47**	**46**	**44**	**42**	**41**	39
B3. Exports	**43**	**50**	**59**	**57**	**57**	**56**	**55**	**53**	**51**	**49**	**47**
B4. Other flows 3/	**43**	**46**	**46**	**44**	**44**	**43**	**42**	**40**	39	37	35
B5. Depreciation	**43**	**70**	**68**	**66**	**65**	**64**	**62**	**59**	**56**	**52**	**49**
B6. Combination of B1-B5	**43**	**49**	**51**	**49**	**49**	**48**	**47**	**45**	**43**	**41**	39
C. Tailored Tests											
C1. Combined contingent liabilities	**43**	**48**	**48**	**47**	**46**	**46**	**45**	**43**	**42**	**40**	38
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**43**	**45**	**45**	**44**	**45**	**44**	**43**	**42**	40	38	36
C4. Market Financing	**43**	**49**	**48**	**47**	**46**	**46**	**44**	**42**	40	37	35
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	125	125	125	120	122	125	126	121	121	120	117
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	125	131	141	148	162	177	**191**	**201**	**220**	**238**	**253**
B. Bound Tests											
B1. Real GDP growth	125	125	125	120	122	125	126	121	121	120	117
B2. Primary balance	125	130	143	139	141	145	146	142	145	145	145
B3. Exports	125	169	**247**	**238**	**243**	**250**	**253**	**247**	**252**	**252**	**252**
B4. Other flows 3/	125	130	135	129	131	134	135	131	132	131	129
B5. Depreciation	125	118	117	112	114	117	117	112	112	110	107
B6. Combination of B1-B5	125	145	138	158	161	165	167	162	164	162	161
C. Tailored Tests											
C1. Combined contingent liabilities	125	137	140	135	138	142	143	139	143	144	143
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	125	126	130	128	132	137	139	135	135	134	132
C4. Market Financing	125	125	126	121	123	126	127	122	121	119	116
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	**19**	**15**	**16**	15	15	**16**	**17**	**19**	**19**	**18**	**19**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	**19**	15	**16**	**16**	**16**	**18**	**20**	**23**	**24**	**24**	**26**
B. Bound Tests											
B1. Real GDP growth	**19**	**15**	**16**	15	15	**16**	**17**	**19**	**19**	**18**	**19**
B2. Primary balance	**19**	**15**	**16**	**16**	**16**	**17**	**18**	**21**	**20**	**19**	**21**
B3. Exports	**19**	**18**	**24**	**26**	**26**	**28**	**29**	**33**	**32**	**31**	**33**
B4. Other flows 3/	**19**	**15**	**16**	**15**	**15**	**16**	**17**	**20**	**19**	**18**	**20**
B5. Depreciation	**19**	**15**	**15**	14	14	**15**	**16**	**19**	**18**	**17**	**18**
B6. Combination of B1-B5	**19**	**17**	**20**	**19**	**19**	**20**	**21**	**25**	**24**	**23**	**24**
C. Tailored Tests											
C1. Combined contingent liabilities	**19**	**15**	**17**	**16**	**16**	**17**	**18**	**21**	**20**	**19**	**21**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**19**	**15**	**16**	**15**	**16**	**17**	**18**	**21**	**20**	**19**	**21**
C4. Market Financing	**19**	**15**	**16**	**15**	**15**	**19**	**21**	**22**	**19**	**17**	**18**
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	**50**	**36**	**37**	**34**	**32**	**34**	**33**	**38**	**34**	**31**	**31**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	**50**	**35**	**37**	**36**	**36**	**39**	**40**	**45**	**43**	**41**	**43**
B. Bound Tests											
B1. Real GDP growth	**50**	**39**	**42**	**39**	**37**	**39**	**38**	**44**	**39**	**35**	**35**
B2. Primary balance	**50**	**36**	**38**	**37**	**36**	**37**	**36**	**41**	**37**	**34**	**34**
B3. Exports	**50**	**36**	**40**	**41**	**40**	**41**	**41**	**45**	**41**	**37**	**38**
B4. Other flows 3/	**50**	**36**	**38**	**35**	**34**	**35**	**35**	**39**	**36**	**32**	**32**
B5. Depreciation	**50**	**62**	**60**	**55**	**53**	**55**	**54**	**62**	**56**	**50**	**50**
B6. Combination of B1-B5	**50**	**38**	**42**	**39**	**37**	**39**	**38**	**43**	**39**	**35**	**36**
C. Tailored Tests											
C1. Combined contingent liabilities	**50**	**36**	**39**	**36**	**35**	**36**	**36**	**41**	**37**	**34**	**34**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**50**	**45**	**47**	**44**	**40**	**40**	**38**	**42**	**38**	**34**	**34**
C4. Market Financing	**50**	**36**	**38**	**35**	**34**	**41**	**42**	**42**	**36**	**29**	**29**
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Ghana: Sensitivity Analysis for Key Indicators of Public Debt , 2020-2030

in percent

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	**68**	**67**	**66**	**65**	**65**	**65**	**62**	**59**	**56**	52	48
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	**68**	**65**	**61**	**58**	**55**	52	49	47	44	42	41
B. Bound Tests											
B1. Real GDP growth	**68**	**72**	**79**	**80**	**82**	**84**	**83**	**82**	**81**	**79**	**78**
B2. Primary balance	**68**	**70**	**77**	**75**	**75**	**75**	**72**	**69**	**66**	**62**	**59**
B3. Exports	**68**	**72**	**83**	**81**	**81**	**81**	**78**	**75**	**72**	**68**	**65**
B4. Other flows 3/	**68**	**68**	**70**	**68**	**68**	**68**	**65**	**62**	**59**	**55**	52
B5. Depreciation	**68**	**95**	**95**	**94**	**94**	**95**	**94**	**93**	**91**	**90**	**89**
B6. Combination of B1-B5	**68**	**72**	**78**	**77**	**78**	**80**	**78**	**76**	**74**	**71**	**69**
C. Tailored Tests											
C1. Combined contingent liabilities	**68**	**76**	**76**	**74**	**74**	**74**	**71**	**68**	**65**	**62**	**58**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**68**	**71**	**76**	**81**	**85**	**90**	**90**	**89**	**88**	**86**	**85**
C4. Market Financing	**68**	**67**	**66**	**65**	**65**	**65**	**62**	**59**	**56**	52	48
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	504	441	441	432	428	426	400	373	347	320	295
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	504	429	407	388	365	343	319	296	278	261	248
B. Bound Tests											
B1. Real GDP growth	504	477	524	529	539	552	536	519	505	488	475
B2. Primary balance	504	466	509	499	494	492	465	437	412	385	360
B3. Exports	504	479	550	540	535	532	504	475	448	419	394
B4. Other flows 3/	504	451	462	453	449	447	420	393	367	339	314
B5. Depreciation	504	627	628	621	621	626	606	586	571	555	543
B6. Combination of B1-B5	504	475	514	513	517	523	502	481	461	440	422
C. Tailored Tests											
C1. Combined contingent liabilities	504	503	502	493	488	487	460	432	407	380	356
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	504	573	616	649	648	644	604	561	547	531	518
C4. Market Financing	504	441	442	434	431	429	403	374	347	319	293
Debt Service-to-Revenue Ratio											
Baseline	107	79	73	80	81	76	76	85	74	63	61
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	107	75	63	66	62	54	50	54	45	38	36
B. Bound Tests											
B1. Real GDP growth	107	85	85	96	100	95	96	109	99	88	89
B2. Primary balance	107	79	80	92	89	83	83	92	85	76	71
B3. Exports	107	79	75	87	89	83	83	93	81	69	68
B4. Other flows 3/	107	79	73	81	83	77	77	87	76	64	63
B5. Depreciation	107	90	97	107	110	109	111	126	118	109	111
B6. Combination of B1-B5	107	82	80	94	95	90	90	101	90	82	80
C. Tailored Tests											
C1. Combined contingent liabilities	107	79	89	88	88	82	82	92	90	73	69
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	107	98	99	114	115	107	103	109	102	94	94
C4. Market Financing	107	79	73	81	82	83	84	90	76	61	60

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.